

Human-Computer Interaction

Qualitative Field Methods

Professor Bilge Mutlu

Today's Agenda

- When to go to the field: nascent theory & fresh looks
- Ethnography: settings, fieldwork, data
- Reading ethnography (Dourish): insight, not requirements
- Probes & diaries: studying life *in the wild*
- Hands-on: *field method for your question*, then a project check-in

Due today, 9:30 am

- **Design-led / WoZ Study Plan** (*individual*)
- **Research Questions + Literature Review** (*team milestone*)

Both on Canvas. Today's worksheet builds on the RQ you just submitted.

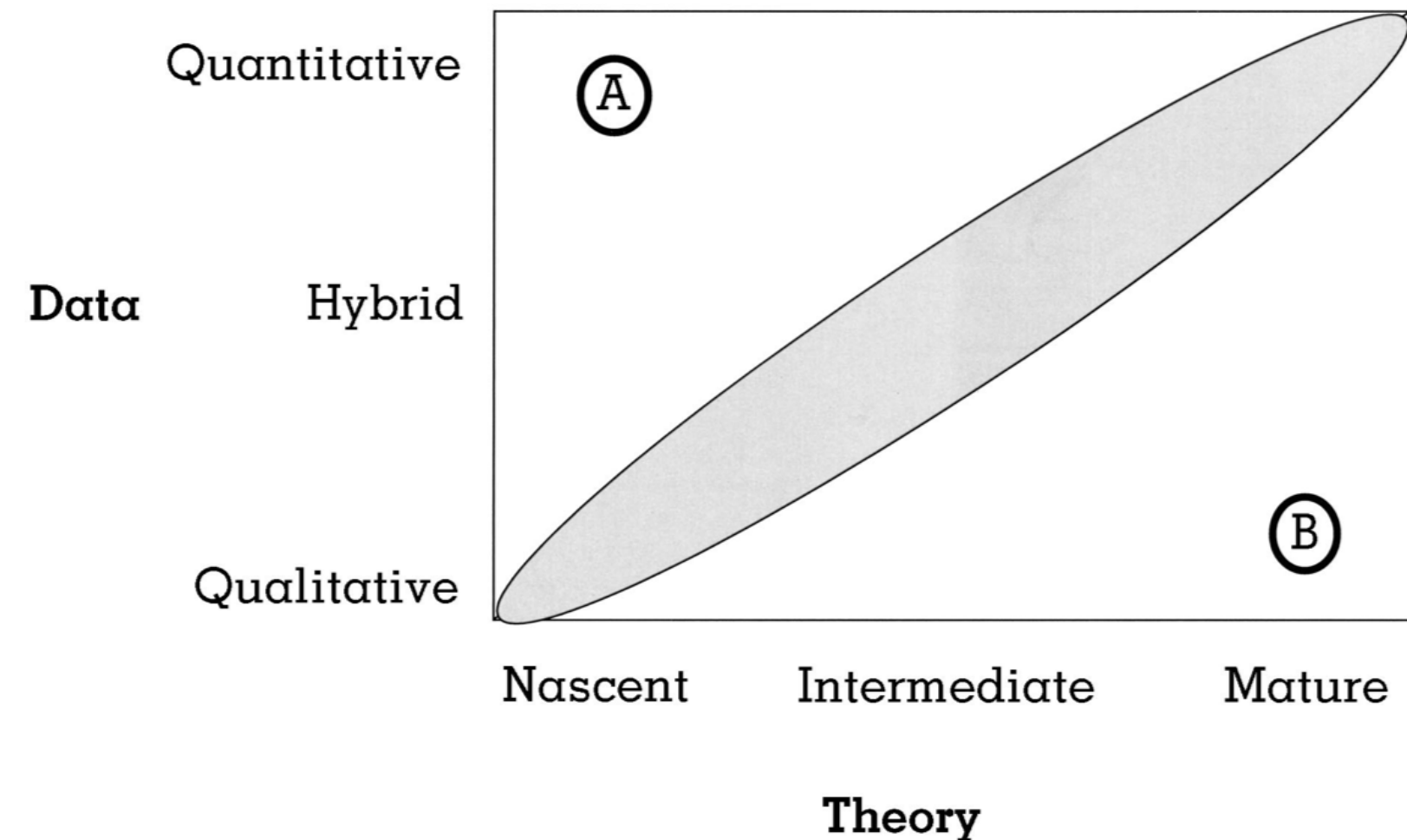
This week's readings

- **Required**
 - Dourish, *Reading and Interpreting Ethnography* (2014)
 - Gaver, Dunne & Pacenti, *Design: Cultural Probes* (1999)
- **Recommended:** LFH **Ch. 9 (Ethnography)** + **Ch. 6 (Diaries)**
- **Optional:** Consolvo & Walker, *Using ESM to Evaluate Ubicomp Applications* (2003)

When do we go to the field?

1. Theory is **nascent** in an area
2. You want to take a **fresh look** at a mature topic

FIGURE 1
Methodological Fit As a Mean Tendency



The qualitative family¹

1. Narrative (inquiry)
2. Phenomenology
3. Grounded theory
4. Case study
5. (Participatory) action research
6. **Ethnography** ← today's focus

¹ Items 1–5: Creswell, J.W., Hanson, W.E., Plano Clark, V.L. & Morales, A. (2007). Qualitative Research Designs: Selection and Implementation. *The Counseling Psychologist*, 35(2), 236–264. Ethnography is the sixth classic tradition.

Ethnography²

Definition: deep immersion and participation in a specific research context to develop an understanding not achievable with more limited approaches.

- Roots in anthropology & sociology, adopted by HCI
- Observation + interview data → new understanding and theory

²Lazar, Feng & Hochheiser (2017). *Research Methods in HCI*, Ch. 9 (Ethnography).

Where do we start?

1. Identify a domain where theory is *nascent* and new theory can have great impact
2. Develop a general *research question* and focus that can be updated
3. Find a *setting* to study the phenomena of interest
4. Conduct *fieldwork*, asking: **"What is going on here?"**

Setting & fieldwork

Setting: a site where the phenomena of interest can be observed.

Fieldwork: an organic process where data *collection* and *analysis* develop symbiotically, becoming more focused over time.

- Obtaining access, informed consent
- Identifying *stakeholders*, choosing a *role*
- Discovering groups & situations for comparison
- Writing detailed *fieldnotes*

How do we collect data?

1. **Fly-on-the-wall observation**: observe without influencing the context
 2. **Participant observation**: understand by taking part in the activities
 3. **Interviews**: understand perspectives; validate observations
- Each produces *fieldnotes* or *transcripts* (written, audio, video).

Reading ethnography³

Dourish: ethnography is *not* a requirements-mining tool.

Read fieldwork for **analytic insight** into how people organize their world, not to extract a feature list.

- The contribution is *understanding*, not a to-do list
- Judge fieldwork on its interpretive frame, not its design payload

³ Dourish, P. (2014). Reading and Interpreting Ethnography. In *Ways of Knowing in HCI*, Springer, pp. 1-23.

Digital ethnography (netnography)⁴

- Extends ethnographic methods to online settings: forums, social media, virtual worlds
- Combines participant observation, interviews, analysis of digital artifacts
- Unique *ethical issues*: consent in online spaces, trace-data use
- Often *multi-sited*: tracing online↔offline practices

⁴Kozinets (2015). *Netnography: Redefined*. Sage. · Boellstorff et al. (2012). *Ethnography and Virtual Worlds*.

*But how do we study life in
the wild, asynchronously and
over time?*

Why probes & diaries?

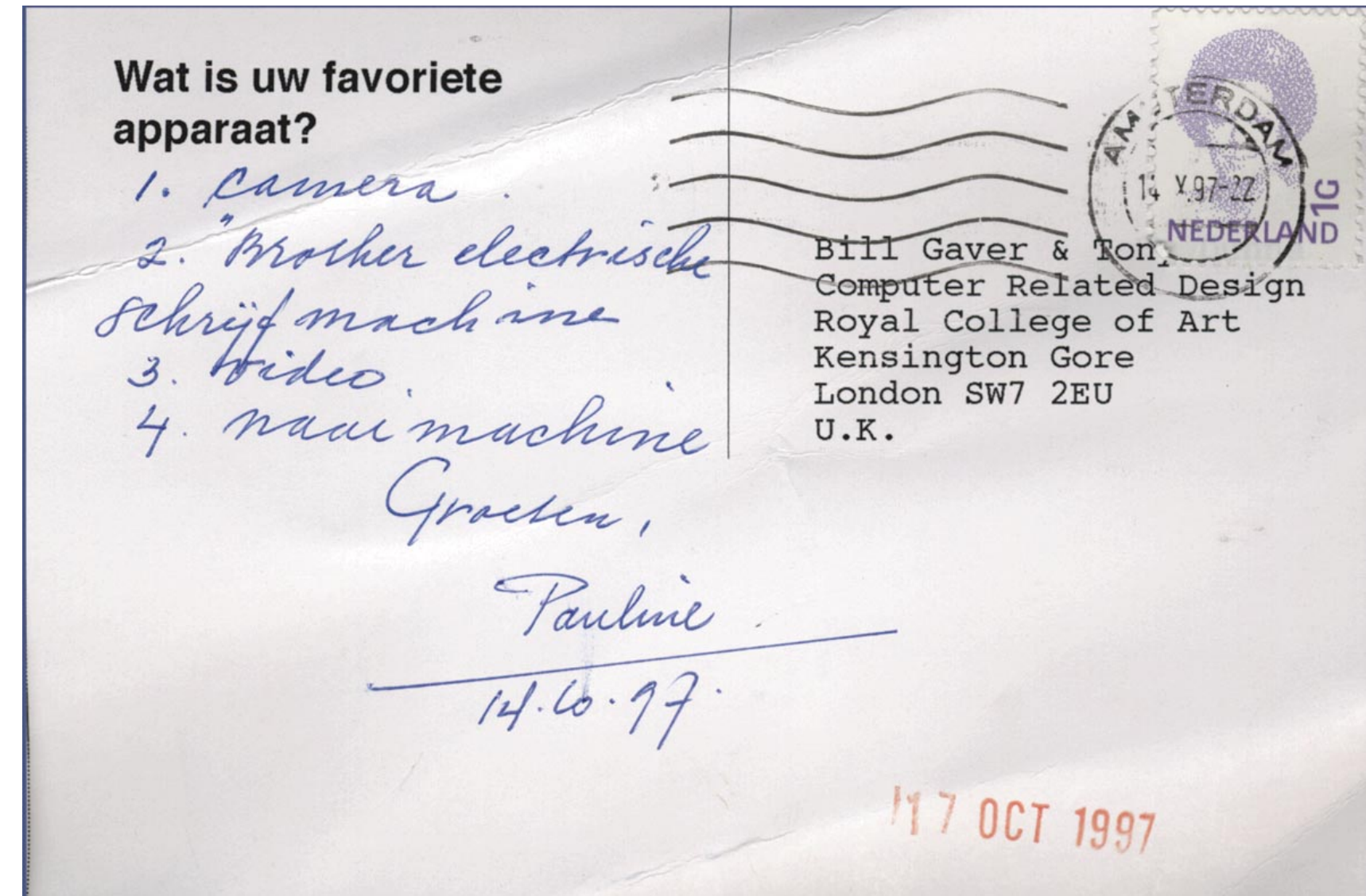
- **Challenge:** study technology use in *naturalistic, everyday contexts* where labs fall short
- **Goal:** capture rich, contextualized, temporal, and personal insight



Cultural probes⁵

Kits of evocative materials left with participants: postcards, maps, a disposable camera, a photo album, a media diary.

- **Inspiration, not information:** provoke design ideas, not measure variables
- Open-ended, playful, aesthetic; center participants' voices & cultural context



⁵ Gaver, Dunne & Pacenti (1999). Design: Cultural Probes. *interactions*, 6(1), 21-29.

Technology probes⁶

Simple technologies deployed in homes, serving three goals at once:

- *Social science*: understand needs in real contexts
- *Engineering*: field-test technology
- *Design*: inspire new ideas

Not hi-fi prototypes: deliberately simple, co-adaptive; collect logs of *actual use*.

⁶ Hutchinson et al. (2003). Technology Probes: Inspiring Design for and with Families. *CHI 2003*.

Diaries & experience sampling

- **Diary:** self-report log of events, feelings, experiences over time
- **Experience Sampling (ESM):** record at *random or prompted* times (beepers → smartphones)

Capture experiences *as they happen* → reduce recall bias; cover multiple contexts.

Choosing the method

Method	Purpose	When to use
Cultural probes	Inspire design ideas	Early design exploration
Technology probes	Explore + co-design	Early/mid design, field testing
Diaries / ESM	Document experiences	Practices & behaviors over time

Decision guide: inspiration → cultural probe · field test + ideas → technology probe · temporal data → diary / ESM

Hands-on activity (30 min)

"Field method for your question"

With your team, take your project and design a qualitative field pass:

- (12 min) Choose. **Ethnographic observation**, a **probe** (cultural or technology), or a **diary / ESM** study for one open part of your RQ, and why it fits
- (18 min) Draft. The actual instrument: observation guide, probe kit, or diary prompts, plus what you'd capture

Grab the worksheet

- One copy per team: File → Make a copy
- Nothing to submit today
- Each of you turns it into your **Qualitative Field-Method Plan**

da.gd/EVHaO



Project check-in (10 min)

Each team, in one breath:

- Your **method family**: field, design-led, or experimental
- **One open design question** you haven't settled

Assignment: Qualitative Field-Method Plan

Due: Fri **Oct 9, 9:30 am** · $\approx$ 1–2 pp, individual, on Canvas

1. **Method choice + rationale:** why it fits your RQ and access
2. **Instrument:** observation guide, probe kit, or diary prompts
3. **Sampling & access:** who, how recruited, over what timescale
4. **Analysis intent:** what you'll do with the data
5. **Reflection:** strengths, weaknesses, what you'd change

Rubric ($\checkmark+$ / $\checkmark$ / $\checkmark-$): completeness · fit · feasibility

What's Next?

- **Monday: Generative AI & Tools for Thought**
 - **Read:** Lee et al. (CHI 2025) · **If you can:** Bush (1945)
 - Post your provocation slide
- **Due Fri Oct 9, 9:30 am:** Field-Method Plan (*individual*)
- **Method design** (draft) → Method-Design Crit, Oct 16